Ghost Directions in Transformer Circuits: Causal Filtering and Compression of Logit Receptors in GPT-2

Shivam Pandey
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Contents

1 Starting Point: Receptor Identification

The model under study is GPT-2 Small (12 layers, 12 heads, $d_{\text{model}} = 768$), evaluated on the gender pronoun (GP) prediction task. The task: given a sentence like "So [Name] is a very organized person, isn't," predict whether the next token is "he" or "she."

The starting point comes from the *Beyond Components* framework. For each attention head (l, h), the OV matrix is $W_{OV} = W_V W_O \in \mathbb{R}^{d_{\text{model}} \times d_{\text{model}}}$. Its SVD gives singular vectors $\{u_k, v_k\}$ with singular values σ_k. The right singular vector $v_k = V_h[k, :]$ is the *write direction* - the direction the head writes into the residual stream. When projected through the unembedding matrix W_U, this direction has an interpretable *logit profile*: the vector $v_k^\top W_U$ tells which tokens are promoted or suppressed.

Binary masks were trained over these singular vectors using the GP dataset (1072 sentences, ×2 with corrupted counterparts = 2144 examples). The mask training identifies which (layer, head, sv_index) triples carry gender signal. The top-ranked directions were inspected by their logit profiles, and three receptors were selected whose pronoun tokens ("he", "she", "him", "her") appeared in the top-10 promoted or suppressed tokens.

A note on experiment numbering. The experiments in this report are numbered non-consecutively (0, 0c, 2, 4, 5, 5b, etc.) because several intermediate experiments were run during the research process but did not produce results worth reporting. The numbering reflects the original lab notebook ordering rather than a cleaned-up sequence.

1.1 The Three Receptors

Name	Host head	SV index	Polarity	Role	Top tokens
R1	L10H9	0	+1	Promotion	His, his, He, him
R2	L11H8	6	+1	(Ghost)	his, His, him
R3	L9H7	1	−1	Inhibition	herself, her, She

Table 1: Three mask-identified receptors. Polarity +1 means the direction's positive projection aligns with "he"; −1 means it aligns with "she." R3's polarity is −1: its positive projection pushes toward female pronouns, so it acts as an inhibitor of the male prediction.

The cosine similarities between write directions are:

$$\cos(v_1, v_2) = +0.566, \quad \cos(v_1, v_3) = -0.775, \quad \cos(v_2, v_3) = -0.464 \tag{1}$$

R1 and R3 are strongly anti-correlated (−0.775). They encode opposing gender signals. Whether this coupling is purely geometric (they happen to point in opposite directions) or reflects genuine computational interaction is addressed in Experiment 2.

1.2 Causal Verification

To confirm causal relevance, single-direction interventions were run following the Beyond Components protocol. For each receptor k, its activation is scaled by a factor s during the forward pass:

$$\hat{a}_k(t) = s \cdot a_k(t) \tag{2}$$

where $a_k(t) = u_k^\top[\text{ctx}(t); 1]$ is the receptor's scalar activation. At scales $s \in \{5, 10, 20, 25\}$, all three receptors produce measurable changes in the he/she logit difference, confirming causal relevance. R1 and R3 produce strong, opposite effects. R2 produces weaker effects, foreshadowing its later identification as a ghost.

2 Experiment 0: Baseline Receptor Diagnostics

Before any interaction experiments, a baseline picture of how receptor signal builds across layers was needed. The *receptor score* at layer ℓ is defined as:

$$s(\ell) = \sum_{k=1}^{3} p_k \cdot g_k(\ell) \tag{3}$$

where $g_k(\ell) = v_k^\top x^{(\ell)}$ is receptor k's activation on the residual stream at layer ℓ, and $p_k \in \{+1, -1\}$ is the polarity. A positive s pushes toward "he"; negative pushes toward "she."

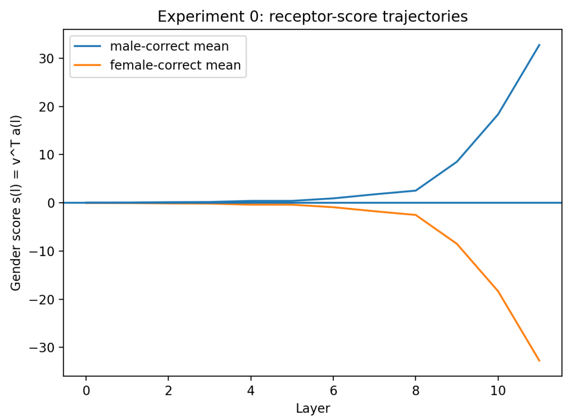
The *Receptor Concentration Score* (RCS) captures how distributed the computation is:

$$\mathrm{RCS}(\ell) = \frac{\max_k |c_k(\ell)|}{\sum_k |c_k(\ell)|} \tag{4}$$

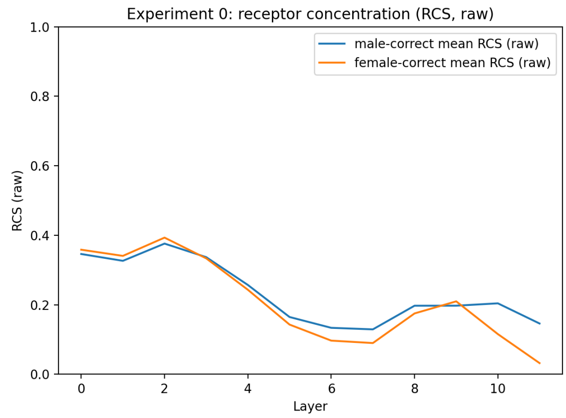
where $c_k(\ell) = p_k \cdot g_k(\ell)$ is receptor k's signed contribution. RCS $= 1$ means one receptor dominates; RCS $= 1/3$ means all three contribute equally.

This was evaluated on 612 examples (306 rows $\times$ 2 with use_both).

2.1 Results



(a) Receptor score trajectories (centered). Male and female means diverge sharply after layer 8.



(b) RCS by gender (raw). Early layers show higher concentration (~ 0.35); late layers show more distributed contributions.

Figure 1: Experiment 0: baseline diagnostics. Left: gender score trajectories separate after layer 8. Right: receptor concentration decreases across layers, indicating a transition from single-receptor dominance to distributed computation.

The trajectory plot (Figure 1a) shows near-zero separation from layers 0–5, then a sharp divergence starting at layer 8–9. By layer 11, male-correct examples reach $s \approx +33$ and female-correct reach $s \approx -33$.

The per-layer AUC rises monotonically:

Layer	0	1	2	3	4	5	6	7	8	9	10	11
AUC	0.58	0.61	0.65	0.66	0.67	0.68	0.78	0.79	0.82	**0.95**	0.96	0.96

Table 2: Per-layer AUC of the combined receptor score $s(\ell)$ for gender classification.

The jump from 0.82 (layer 8) to 0.95 (layer 9) coincides with R3's host head L9H7. This is the first indication that layer 9 is where the circuit becomes operational.

Baseline model accuracy: 91.3% (he-vs-she only), with 15.4% of examples having a non-pronoun top-1 prediction.

3 Experiment 0c: Commitment Analysis

The next question was *when* each receptor commits to its final sign. The commitment layer is the earliest layer ℓ where receptor k's sign agrees with its final-layer sign for $\geq 85\%$ of examples, and this agreement holds at all subsequent layers.

Formally, for receptor k at layer ℓ:

$$\text{agree}_k(\ell) = \frac{1}{N}\sum_{i=1}^{N} \mathbf{1}\left[\text{sign}(g_k^{(\ell)}(i)) = \text{sign}(g_k^{(11)}(i))\right] \tag{5}$$

The commitment layer is $L_k^* = \min\{\ell : \text{agree}_k(\ell') \geq 0.85 \ \forall \ \ell' \geq \ell\}$.

Three other diagnostics were tracked per layer:

- **AUC**: ROC-AUC of $p_k \cdot g_k(\ell)$ for gender classification.
- **Delta**: mean $|\Delta g_k| = |g_k(\ell+1) - g_k(\ell)|$, measuring where signal is written.
- **Deliberation Index**: within-class variance of $g_k(\ell)$, normalized by layer -1.

3.1 Results

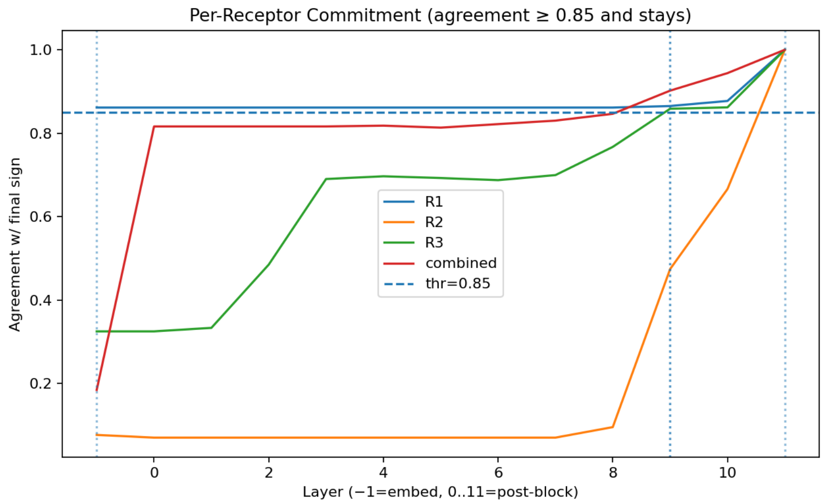
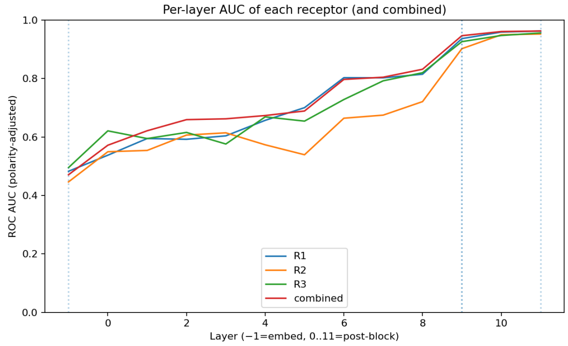


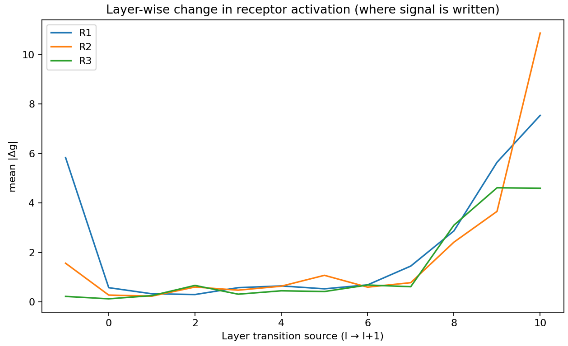
Figure 2: Per-receptor commitment curves. R1 (blue) is committed from the embedding layer ($\ell = -1$). R3 (green) commits at layer 9. R2 (orange) commits only at layer 10–11. The dashed line marks the 0.85 threshold.

The key finding: **R1 never has to "commit" because it is committed from the start.** Its sign agreement exceeds 0.85 from the embedding layer itself. R1 inherits its gender direction from the token embedding - it does not compute it. The name "Brooks" already projects positively onto R1's write direction before any attention head fires.

R3, by contrast, crosses 0.85 at layer 9, exactly where its host head L9H7 sits. R3 is a *computed* signal. R2 commits even later (layer 10–11), consistent with its host head L11H8.



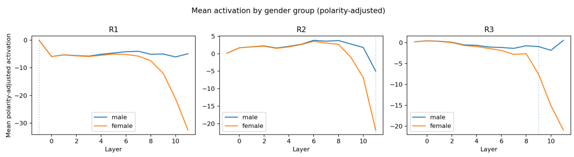
(a) Per-layer AUC. All receptors cross 0.90 at layer 9.



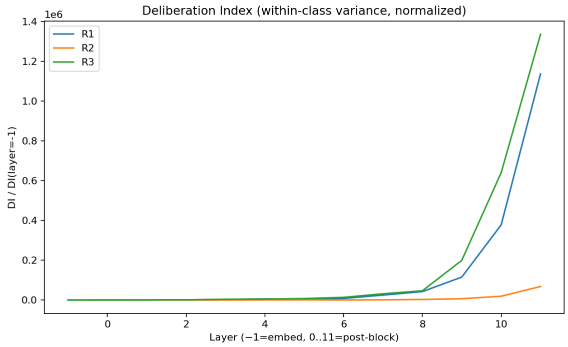
(b) Layer-wise $|\Delta g|$. R3 peaks at $9 \to 10$; R1 and R2 peak at $10 \to 11$.

Figure 3: AUC trajectories and delta peaks from Experiment 0c.

The delta peaks (Figure 3b) serve as fingerprints: R3's signal is written at the $9 \to 10$ transition by its host head L9H7. R1's signal is amplified at $10 \to 11$ by L10H9. R2's signal appears at $10 \to 11$ as well, written by L11H8.



(a) Mean polarity-adjusted activation by gender group. R1 shows separation from the embedding. R3 separates at layer 9.



(b) Deliberation index (within-class variance, normalized). Reaches $10^6\times$ the embedding variance at layers 9–11.

Figure 4: Activation by gender and deliberation from Experiment 0c.

The deliberation explosion (Figure 4b) is notable. Within-class variance grows by a factor of $\sim 10^6$ from the embedding to the final layer. Late layers do not just separate genders - they amplify individual differences within each gender class. This amplification creates both the strong correct predictions (large margins) and the failures on ambiguous names (names that happen to get amplified in the wrong direction).

Interpretation. The circuit has an *inherited-vs-computed asymmetry*. R1's signal is free: the embedding already contains gender information along R1's direction. R3's signal is computed: L9H7 must actively write it. This asymmetry has consequences for robustness - R1 is reliable but coarse, R3 is powerful but fragile.

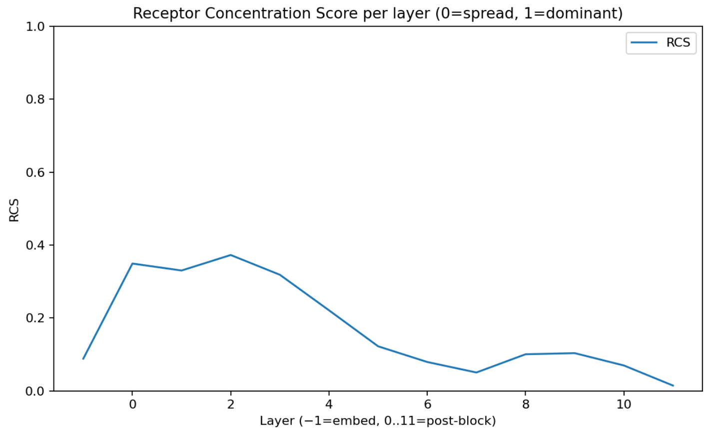


Figure 5: Receptor Concentration Score per layer (from Experiment 0c). RCS peaks around layers 0–2 ($\sim$0.37) and drops toward zero at layer 11. The circuit moves from single-receptor dominance to distributed computation.

4 Experiment 2: Cross-Talk Under Scaling

R1 and R3 have cosine similarity -0.775. Scaling R3's activation changes R1's activation. But is this change purely geometric (because v_1 and v_3 are not orthogonal) or does it reflect genuine computational coupling (R3's perturbation propagates through intermediate layers and changes R1's output)?

To disentangle these two effects: when receptor k is scaled by factor s, receptor j's activation changes by Δg_j. The geometric prediction is:

$$\Delta g_j^{\text{geo}} = (s-1) \cdot \sigma_k \cdot \bar{a}_k \cdot \cos(v_k, v_j) \tag{6}$$

This is what would happen if the perturbation passed through without any nonlinear processing. The *computational fraction* is:

$$\text{CompFrac}(k \to j) = 1 - \frac{|\Delta g_j^{\text{geo}}|}{|\Delta g_j^{\text{actual}}|} \tag{7}$$

CompFrac $= 0$ means all cross-talk is geometric. CompFrac > 0 means there is computational coupling beyond geometry.

The experiment was run at scales $s \in \{0.1, 0.5, 1.0, 2.0, 5.0, 10.0, 20.0\}$, measuring cross-talk at each.

4.1 Results

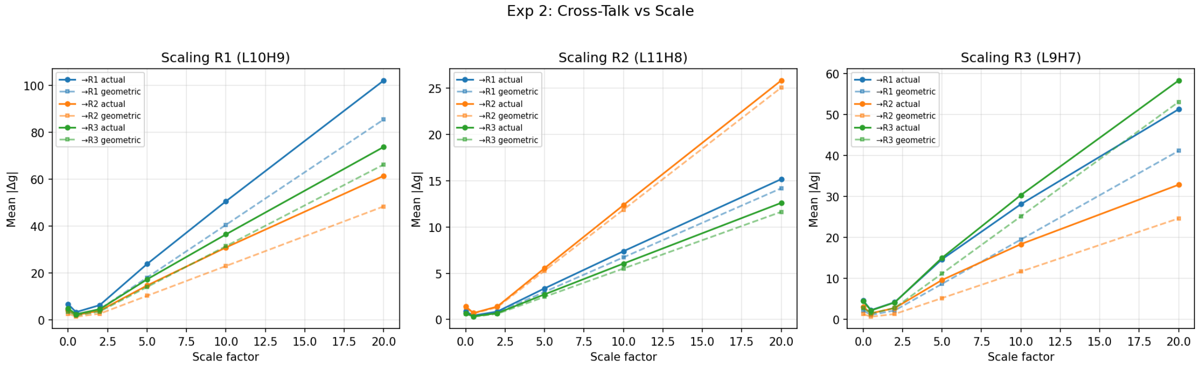
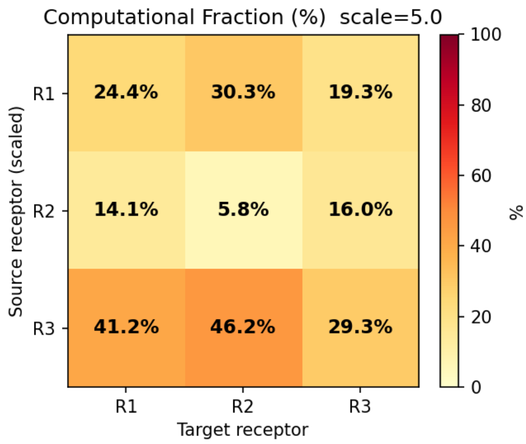
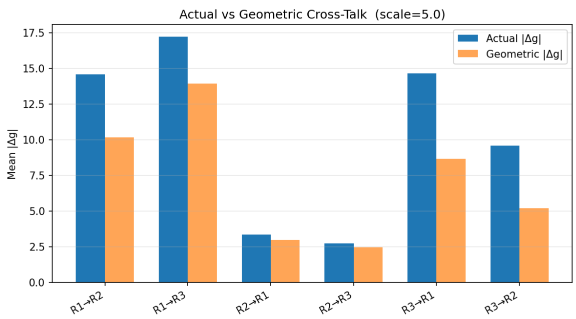


Figure 6: Cross-talk vs. scale for all source-target pairs. Solid lines: actual $|\Delta g|$. Dashed lines: geometric prediction. The gap between solid and dashed is the computational component.



(a) Computational fraction matrix at scale 5.0.



(b) Actual vs. geometric $|\Delta g|$ at scale 5.0.

Figure 7: Experiment 2: cross-talk decomposition.

The computational fraction matrix (Figure 7a) reveals clear structure:

Source $\rightarrow$ Target	R1	R2	R3
R1 $\rightarrow$	24.4%	30.3%	19.3%
R2 $\rightarrow$	14.1%	5.8%	16.0%
R3 $\rightarrow$	**41.2%**	**46.2%**	29.3%

Table 3: Computational fraction at scale 5.0. R3 $\rightarrow$ R1 coupling is 41.2% - nearly half the effect is computational.

Three findings stand out.

R2's self-CompFrac is 5.8%. R2's host head L11H8 sits in the last layer; after it writes, only ln_final remains. The perturbation passes through almost unchanged. This serves as a clean control: the methodology works.

R3 $\rightarrow$ R1 CompFrac is 41.2%. When R3 (L9H7) is scaled, 41% of the resulting change in R1 cannot be explained by geometry. The perturbation at layer 9 passes through layers 10 and 11, and L10H9 (R1's host head) reads the modified residual stream and recomputes. This is genuine computational coupling.

The asymmetry is directional. R3 → R1 CompFrac (41.2%) exceeds R1 → R3 CompFrac (19.3%). This follows from circuit architecture: R3's host head fires before R1's, so R3's perturbation flows *through* R1's head, but not vice versa. This asymmetry was predicted before running the experiment; the data confirms it.

Linearity check: R^2 of mean $|\Delta g|$ vs. $|s - 1|$ exceeds 0.99 for all pairs, confirming the perturbation regime is linear.

5 Experiment 4: Shapley Ablation and Independence

To quantify each receptor's individual contribution, Shapley values provide a principled decomposition. For each subset $S \subseteq \{R1, R2, R3\}$, the directions *not* in S are projected out from the final residual stream, and logits are recomputed via ln_final and W_U.

The Shapley value for receptor k on metric v (accuracy or mean margin) is:

$$\phi_k = \frac{1}{K!} \sum_{\pi} \left[v(S_\pi^k \cup \{k\}) - v(S_\pi^k) \right] \tag{8}$$

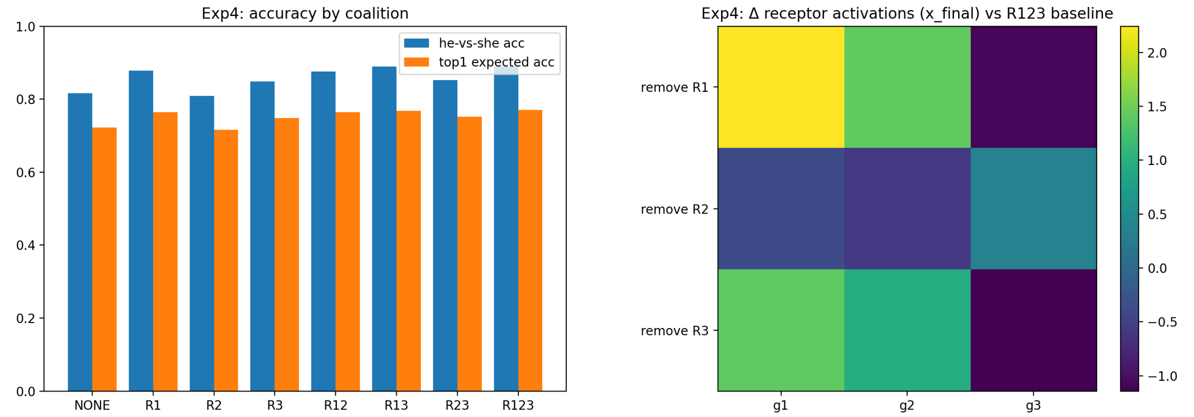
where the sum is over all orderings π of the $K = 3$ receptors, and S_π^k is the set of receptors preceding k in ordering π.

With $K = 3$, there are $2^3 = 8$ coalitions. All eight were evaluated.

5.1 Results

Coalition	Acc (he-vs-she)	Top-1 Acc	Mean margin	RCS
NONE (all ablated)	0.816	0.722	+0.975	0.031
R1 only	0.879	0.764	+1.985	0.023
R2 only	0.809	0.716	+1.065	0.031
R3 only	0.849	0.748	+1.537	0.021
R1+R2	0.876	0.765	+2.103	0.022
R1+R3	0.890	0.769	+2.450	0.015
R2+R3	0.853	0.752	+1.643	0.020
R1+R2+R3 (baseline)	0.889	0.771	+2.579	0.015

Table 4: Coalition accuracies and margins from Experiment 4.



(a) Accuracy by coalition.

(b) Δ activations when removing each receptor.

Figure 8: Experiment 4: Shapley ablation results.

The Shapley values for accuracy: $\phi_{\text{acc}}(R1) = +0.051$, $\phi_{\text{acc}}(R2) = -0.003$, $\phi_{\text{acc}}(R3) = +0.024$. For mean margin: $\phi_{\text{margin}}(R1) = +0.974$, $\phi_{\text{margin}}(R2) = +0.110$, $\phi_{\text{margin}}(R3) = +0.520$.

R2 contributes nothing. Its accuracy Shapley value is -0.003 (removing it actually helps slightly), and its margin Shapley is only 0.110 out of 1.604 total. This was the first quantitative evidence that R2 is a ghost - a direction that appears gender-aligned in logit space but contributes negligibly to the task.

R1 dominates margin. R1's margin Shapley (0.974) is nearly twice R3's (0.520). R1 is the primary driver of confident predictions.

The interaction indices confirm anti-cooperative behavior between R1 and R3: $I_{13}(\text{acc}) = -0.011$, $I_{13}(\text{margin}) = -0.048$. Having both provides slightly less than the sum of their individual contributions, consistent with their anti-correlated directions partially canceling.

6 Phase Diagram and Error Autopsy

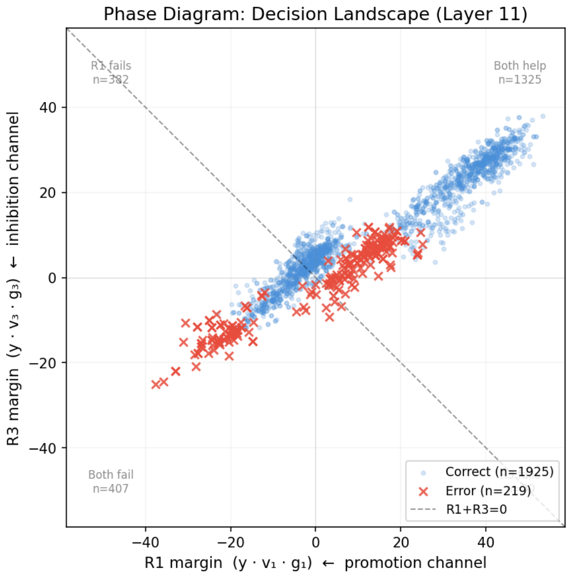
To map the decision landscape, for each example i, the *R1 margin* is defined as $m_1(i) = y_i \cdot p_1 \cdot g_1^{(11)}(i)$ and the *R3 margin* as $m_3(i) = y_i \cdot p_3 \cdot g_3^{(11)}(i)$, where $y_i \in \{+1, -1\}$ is the label. Positive margin means the receptor votes correctly.

Plotting m_1 vs. m_3 for all examples gives a phase diagram. Errors fall into three types:

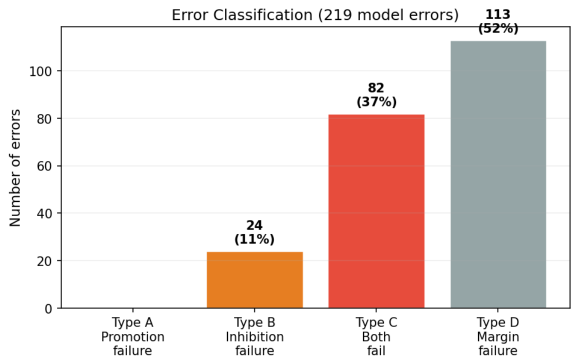
- **Type B (inhibition failure):** R1 correct ($m_1 > 0$), R3 wrong ($m_3 < 0$). R3 failed to suppress.
- **Type C (both fail):** $m_1 < 0$ and $m_3 < 0$.
- **Type D (margin failure):** Both correct ($m_1 > 0$, $m_3 > 0$), but combined margin too small; other logit components overwhelm.

(Type A - promotion-only failure with $m_1 < 0$ and $m_3 > 0$ - did not occur in the data.)

6.1 Results



(a) Phase diagram at final layer. Blue dots: correct. Red crosses: errors. Errors cluster near the origin (low margins).



(b) Error classification. 219 total errors: Type B = 24 (11%), Type C = 82 (37%), Type D = 113 (52%).

Figure 9: Phase diagram and error autopsy.

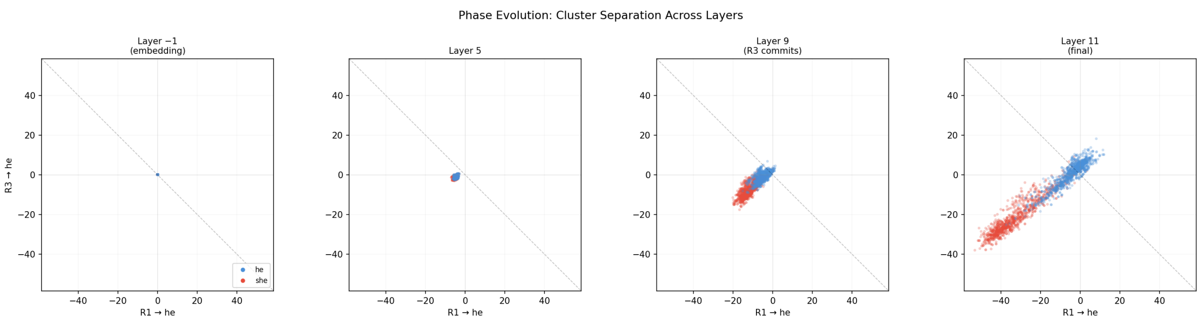
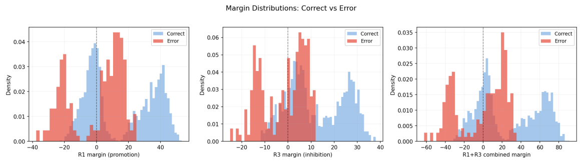


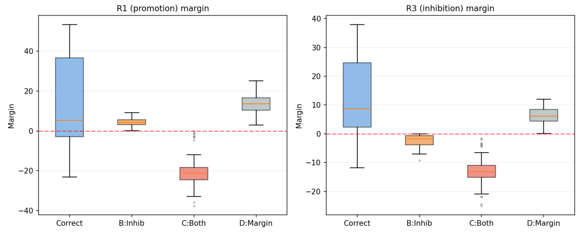
Figure 10: Phase evolution across layers. At layer −1 (embedding), all points are clustered near the origin. By layer 9, gender clusters emerge. By layer 11, full separation.

The error breakdown was unexpected. The prior expectation was that Type B (inhibition failure) would dominate, because R3 is the computed, fragile channel. Instead, **Type D (margin failure) is the majority at 52%**. In these 113 cases, both R1 and R3 vote correctly, but their combined margin is small enough that other logit components push the prediction the wrong way.

Type C (both fail, 37%) corresponds to genuinely ambiguous names - cases where the name's embedding already points the wrong direction on both R1 and R3.



(a) Margin distributions for correct vs. error examples.



(b) Per-receptor margins by error type.

Figure 11: Margin analysis. Errors are concentrated near zero margin. Type C errors show strongly negative margins on both receptors. Type D errors show positive but small margins.

The margin distributions (Figure 11a) confirm that error examples cluster near zero combined margin. Circuit coverage (Figure 12) shows that roughly 35% of model errors have negative R1+R3 margin across all layers, dropping from 65% at the embedding.

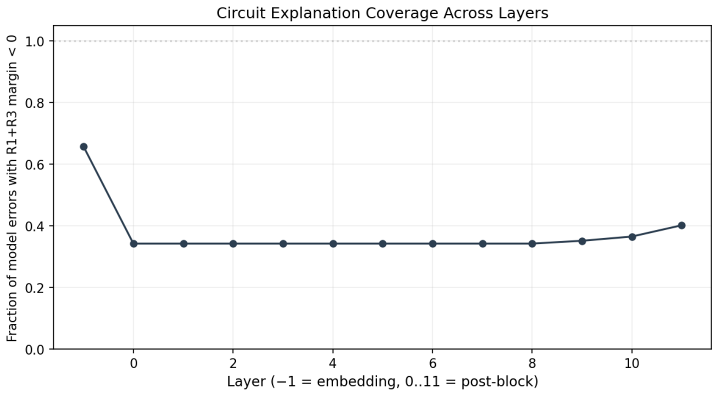


Figure 12: Circuit explanation coverage across layers. Fraction of model errors where R1+R3 margin is negative. Drops from 0.65 at the embedding to ∼0.35 at later layers.

7 Type D Gap Analysis

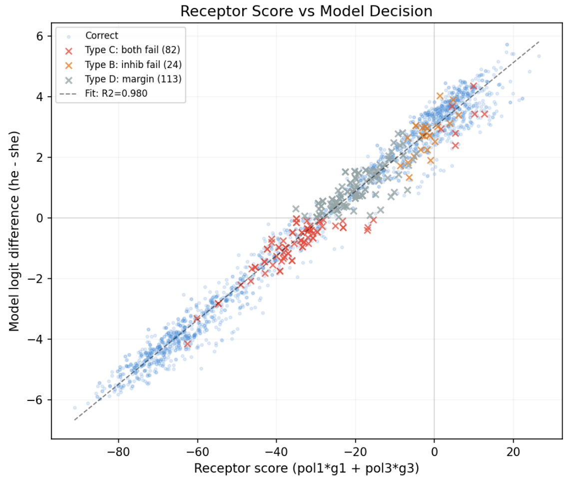
The 113 Type D errors raise a natural question: if both receptors vote correctly, *what* is overriding them? The model's logit difference decomposes into a receptor component and a residual:

$$\underbrace{\mathrm{logit(he)} - \mathrm{logit(she)}}_{\text{model decision}} = \underbrace{(v_1^\top x) \cdot w_1 + (v_3^\top x) \cdot w_3}_{\text{receptor subspace}} + \underbrace{\text{remainder}}_{\text{other components}} \tag{9}$$

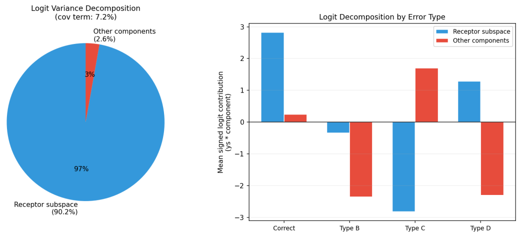
where w_k are the projections of the unembedding difference vector onto the receptor directions.

More precisely, the final residual stream $x^{(11)}$ is projected onto the 3-receptor subspace $\mathrm{span}(v_1, v_2, v_3)$ and its orthogonal complement. The receptor score is $\hat{s} = p_1 g_1 + p_3 g_3$ (ignoring R2 since $\phi(R2) \approx 0$).

7.1 Results



(a) Receptor score vs. model logit difference. $R^2 = 0.980$. Error types are color-coded.



(b) Variance decomposition. Receptor subspace explains 90.2% of logit variance.

Figure 13: Type D gap analysis.

The two-receptor model explains 98.0% of the variance in the model's logit difference ($R^2 = 0.980$). The receptor subspace accounts for 90.2% of total logit variance; the orthogonal complement accounts for 2.6%; the covariance term is 7.2%.

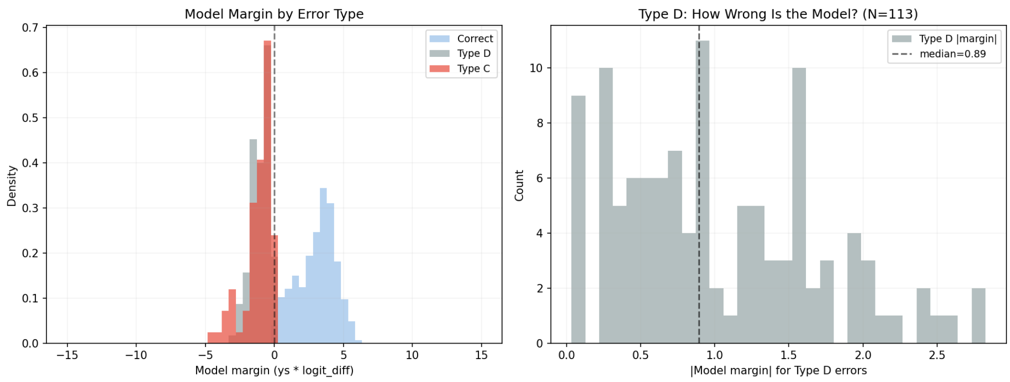


Figure 14: Left: model margin by error type. Type D errors cluster near zero. Right: absolute model margin for Type D errors. Median is 0.89 - these are razor-thin losses.

For the 113 Type D errors, the median absolute model margin is 0.89 logit units. These are cases where the receptor circuit gives the right answer, but the 2.6% residual (other logit components) tips the balance. For correct examples, the receptor subspace contributes positively and the residual is small. For Type C errors, the receptor subspace itself is negative. For Type D errors, the receptor subspace is positive but small, and the residual pushes negative.

This decomposition explains why Type D is the dominant error mode. The receptor circuit has 98% explanatory power, which sounds near-perfect, but that remaining 2% is enough to flip predictions whenever the receptor margin is thin. The circuit is accurate in direction but occasionally insufficient in magnitude.

8 Experiment 5: Cross-Structure Generalization

All experiments so far used one sentence structure: "So [Name] is a very [adj] person, isn't __". To test whether the receptor directions generalize across syntactic contexts, five sentence structures were constructed:

- **A: Tag question** (baseline): "So [Name] is a very [adj] person, isn't __"
- **B: Because**: "Because [Name] is [adj], __"
- **C: Temporal**: "After [Name] finished the project, __"
- **D: Prep-of**: "A friend of [Name] said that __"
- **E: Cross-sentence**: "[Name] is [adj]. __"

For each structure, ∼1200 examples were generated using the same name pool, a forward pass was run, and receptor AUC was measured at the decision position.

8.1 Results

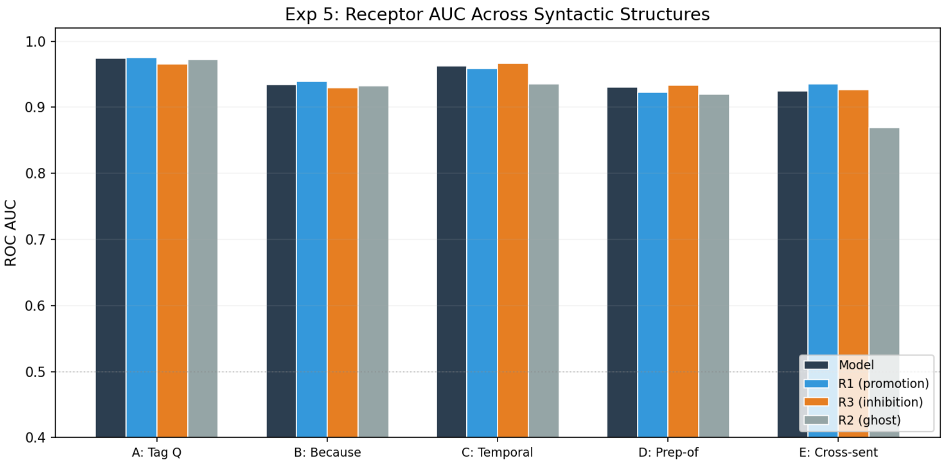


Figure 15: Receptor AUC across five syntactic structures. All remain above 0.92 for R1 and R3. R2 (ghost) drops to 0.87 on cross-sentence.

All structures maintain AUC ≥ 0.92 **for R1, R3, and the model.** The generalization gap (Figure **??**) shows drops of at most 5 percentage points from the baseline structure. The largest drop is R1 on structure D (Prep-of), falling by ∼0.05.

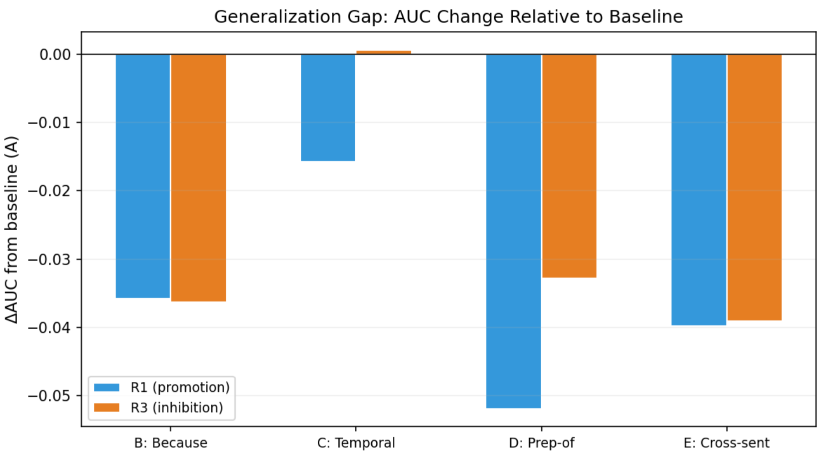


Figure 16: AUC drop relative to baseline (structure A). Maximum drop is ∼0.05 (R1 on Prep-of). R3 is the most stable across structures.

The receptor directions generalize robustly. Even on cross-sentence structure (E), where the period token might reset attention patterns, AUC remains above 0.92 for both R1 and R3.

9 Experiment 5b: Ambiguous Names

Names from the dataset that appear in both male and female contexts have a *he-fraction* $\in (0, 1)$. Names with he-fraction near 0.5 are ambiguous (e.g., Angel, Tyler, Christian). Receptor activations and model predictions were measured for 128 ambiguous examples and compared to 571 unambiguous examples.

9.1 Results

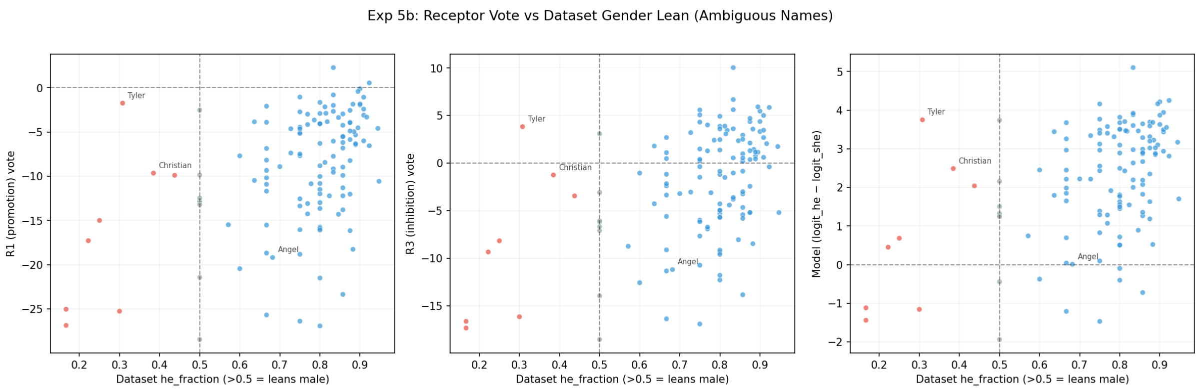


Figure 17: Receptor vote vs. dataset gender lean for ambiguous names. Left: R1 (promotion). Middle: R3 (inhibition). Right: model logit difference. Ambiguous names (gray, near $x = 0.5$) cluster near the decision boundary. Tyler, Christian, and Angel are highlighted.

AUC drops from ∼0.97 (unambiguous) to ∼0.75 (ambiguous) for all receptors and the model alike. The drop is uniform across R1, R3, R2, and the model. This means the performance loss is not a failure of any specific receptor - it is a property of the input. Ambiguous names project weakly onto both R1 and R3, producing small margins that are easily overwhelmed.

Tyler, Christian, and Angel are among the most ambiguous: their receptor activations sit near the decision boundary regardless of the intended gender context.

10 Causal Fidelity Ratio: Ghost Detection

The mask training identified 207 (layer, head, sv_index) triples with nonzero mask weights. But Experiment 4 showed that R2 contributes nothing despite appearing gender-aligned. How many of the 207 are similarly hollow?

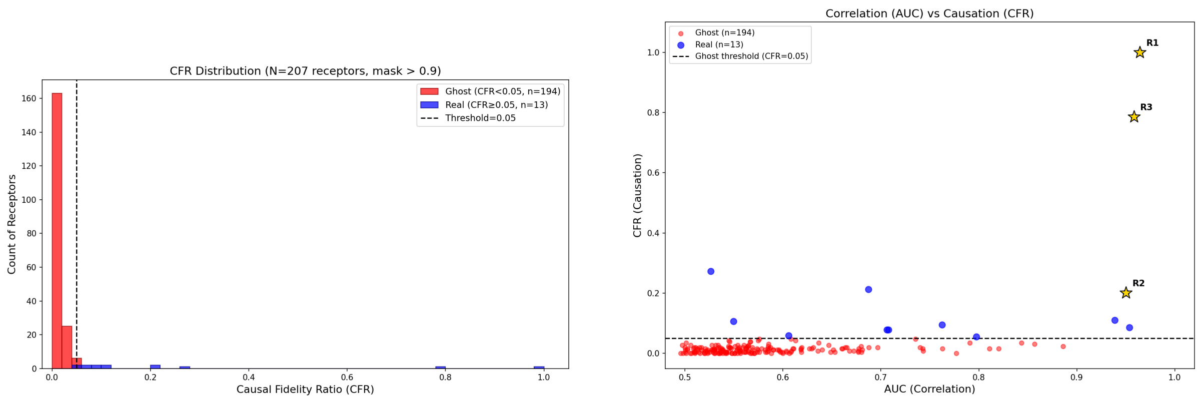
The *Causal Fidelity Ratio* (CFR) tests each direction:

$$\mathrm{CFR}_k = \frac{\mathrm{AccDrop}(k)}{\mathrm{AUC}_k - 0.5} \tag{10}$$

where AccDrop(k) is the accuracy drop when receptor k's direction is projected out of the final residual stream, and AUC_k measures how well the direction separates he-contexts from she-contexts. A direction with high AUC but zero AccDrop is a ghost: it "looks" gendered but removing it does not hurt. CFR captures this: ghosts have CFR ≈ 0; real receptors have CFR $\gg 0$.

All 207 directions were tested on the 612-example test set, with a batch size of 8 and one forward pass per direction.

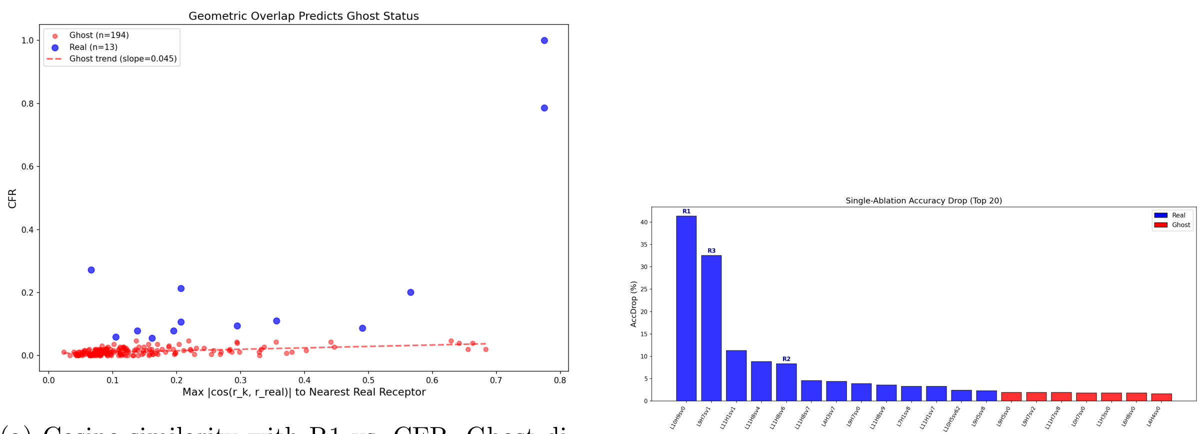
10.1 Results



(a) CFR histogram. 194 of 207 directions have CFR < 0.1.

(b) AUC vs. CFR. Many high-AUC directions have zero CFR.

Figure 18: CFR ghost detection results.



(a) Cosine similarity with R1 vs. CFR. Ghost directions are often highly aligned with R1.

(b) Top 20 directions by AccDrop.

Figure 19: CFR analysis continued.

194 out of 207 directions are ghosts (CFR < 0.1). The ghost rate is 93.7%. Only 13 directions survive. The mask training vastly over-identifies directions as relevant. The majority are *geometric ghosts* - they have high AUC because they are correlated with a real receptor's direction ($\cos > 0.5$ with R1 or R3), not because they carry independent signal.

The cosine plot (Figure 19a) confirms this: ghost directions cluster at high cosine similarity with R1. They are echoes, not independent signals.

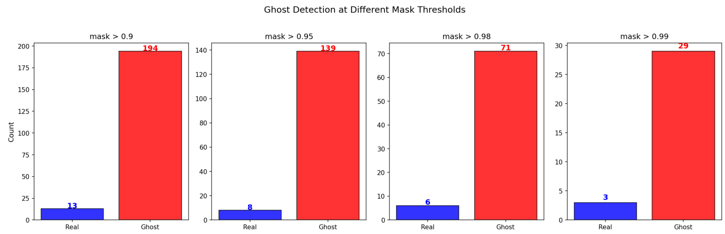


Figure 20: Survivor count at different CFR thresholds.

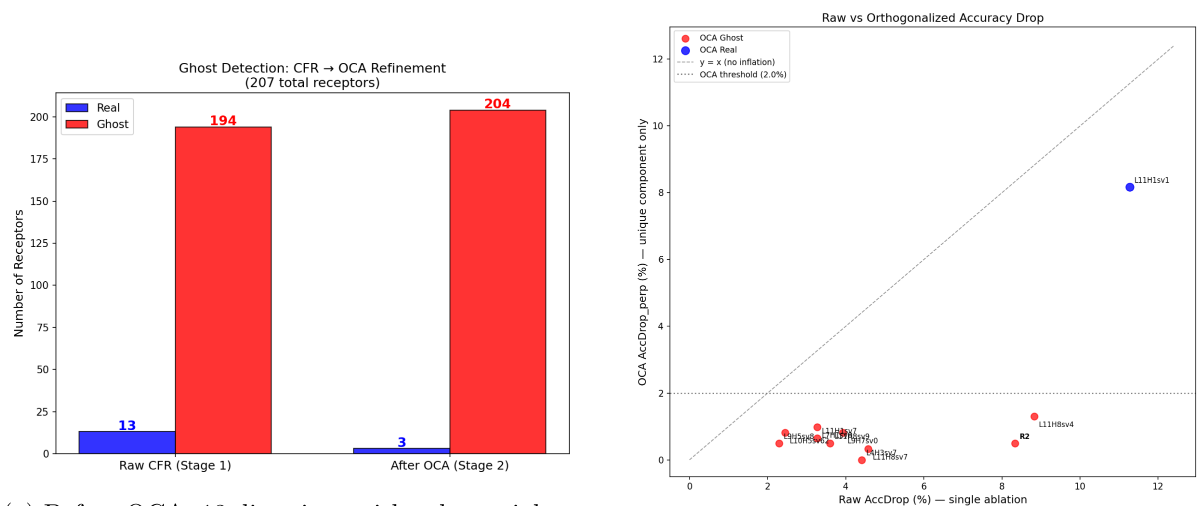
11 OCA: Orthogonal Component Analysis

CFR reduced 207 to 13 survivors. But are these 13 truly independent, or are some still redundant? *Orthogonal Component Analysis* (OCA) addresses this by iteratively pruning directions that are linearly explained by the current basis.

The algorithm:

1. Sort the 13 CFR survivors by AccDrop (highest first).
2. Initialize the basis $\mathcal{B} = \emptyset$.
3. For each candidate direction v, compute its *independence score*: $\text{Indep}(v) = 1 - \max_{b \in \mathcal{B}} |\cos(v, b)|$. If $\text{Indep}(v) < \tau$ (threshold 0.3), the direction is a *correlation ghost* and is pruned.
4. Otherwise, add v to $\mathcal{B}$.
5. After all candidates are processed, measure total R^2: regress the model's logit difference on the activations of the surviving directions.

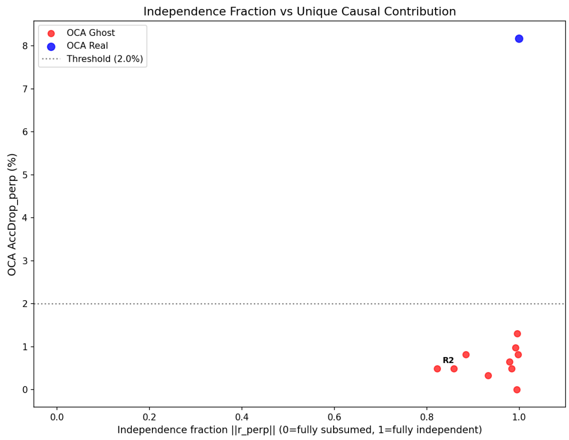
11.1 Results



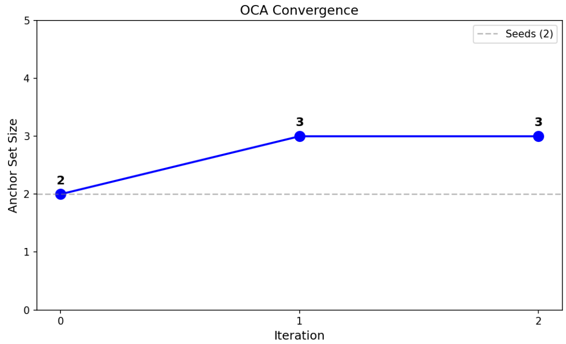
(a) Before OCA: 13 directions with substantial redundancy. After: 3 independent directions.

(b) Raw AccDrop vs. OCA-filtered AccDrop.

Figure 21: OCA refinement: 13 $\rightarrow$ 3.



(a) Independence score vs. AccDrop. Pruned directions (red) have low independence.



(b) R^2 convergence. Three directions achieve $R^2 = 0.9815$.

Figure 22: OCA independence analysis and convergence.

OCA prunes the 13 CFR survivors to exactly 3 independent directions:

- R1: L10H9sv0 (promotion, AccDrop = 3.59%)
- R3: L9H7sv1 (inhibition, AccDrop = 1.31%)
- **Dark receptor**: L11H1sv1 (AccDrop = 8.17%, AUC = 0.526)

The combined $R^2 = 0.9815$. Three directions explain 98.15% of the variance in the model's gender logit difference. The full pipeline - mask training (207) → CFR (13) → OCA (3) - achieves a 69× compression.

The dark receptor is unexpected. It has the *highest* AccDrop (8.17%) but the *lowest* AUC (0.526, barely above chance for gender classification). Removing it hurts accuracy substantially, yet it does not discriminate between he-contexts and she-contexts. It contributes to the logit variance without encoding gender at the pronoun level.

12 Dark Receptor Characterization and PRM

The dark receptor (L11H1sv1) was characterized along three axes: its logit-space profile, its activation distribution by gender context, and its ability to predict model errors.

12.1 Token Profile

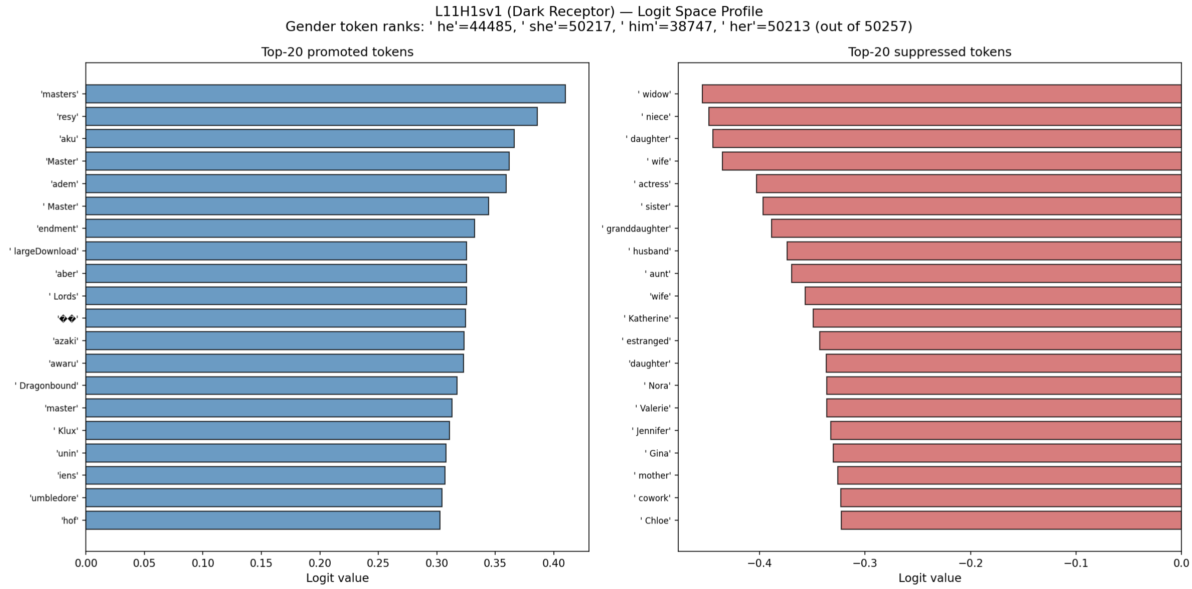


Figure 23: Dark receptor logit profile. Top promoted tokens: "masters", "Master", "Lords". Top suppressed tokens: "widow", "niece", "daughter", "wife". Pronoun tokens rank near the bottom ("he" at rank 44485, "she" at rank 50217 out of 50257).

The dark receptor encodes a *relational-role axis*. It suppresses kinship and relational terms (widow, niece, daughter, wife, sister, granddaughter) and promotes authority/title terms (masters, Master, Lords). The actual pronouns (he, she, him, her) rank near the very bottom of the 50257-token vocabulary. This direction is orthogonal to the pronoun-gender axis in logit space.

12.2 Activation Distributions

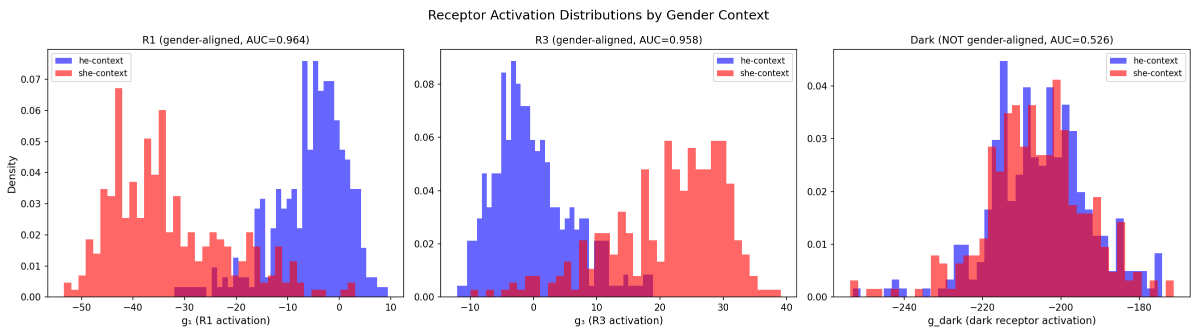


Figure 24: Receptor activation distributions by gender context. R1 (AUC = 0.964) and R3 (AUC = 0.958) show clean separation. The dark receptor (AUC = 0.526) shows complete overlap.

The dark receptor's activations are nearly constant: mean = −205.7, std = 12.7 (coefficient of variation 6.2%). The he-context mean (−205.1) and she-context mean (−206.4) differ by only $d' = 0.10$. The direction matters for the logit computation (removing it drops accuracy by 8.17%), but its activation barely varies across examples. This points to structural infrastructure - a large constant offset that maintains the global geometry of the residual stream - rather than a per-example discriminative feature.

12.3 Predictive Receptor Margin: A Null Result

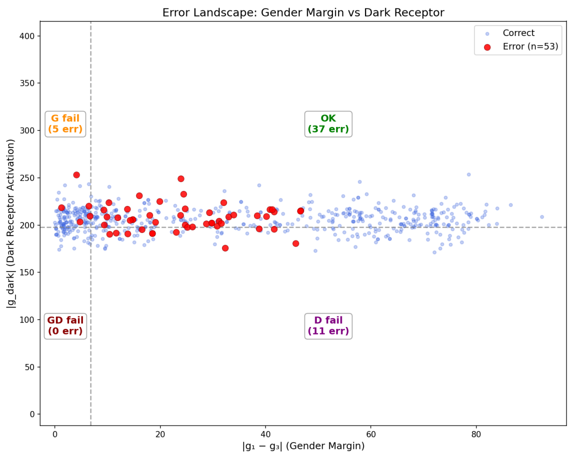
The hypothesis was that the dark receptor might function as a routing or gating signal. If the model needs both gender information (R1, R3) *and* a routing signal (dark receptor) to predict correctly, then errors should cluster where the gender margin is present but the dark signal is weak. Three predictors were defined:

$$\mathrm{PRM}_2 = |g_1 - g_3| \quad \text{(gender margin only)} \tag{11}$$
$$\mathrm{PRM}_3 = |g_1 - g_3| \times |g_{\mathrm{dark}}| \quad \text{(gender} \times \text{dark)} \tag{12}$$
$$\mathrm{PRM}_{\mathrm{dark}} = |g_{\mathrm{dark}}| \quad \text{(dark only)} \tag{13}$$

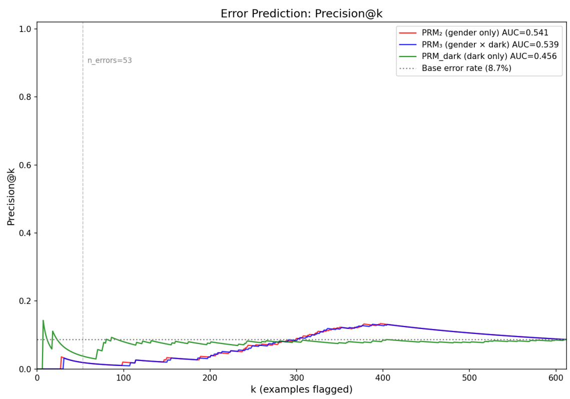
These were evaluated for AUC-ROC on predicting model errors (53 errors out of 612 examples, 8.7% base rate).

Predictor	AUC-ROC	Precision@53
PRM_2 (gender only)	0.541	1.9%
PRM_3 (gender × dark)	0.539	1.9%
$\mathrm{PRM}_{\mathrm{dark}}$ (dark only)	0.544	3.8%
Random baseline	0.500	8.7%

Table 5: Error prediction performance. All predictors perform at chance. PRM_3 is worse than PRM_2 - adding the dark receptor hurts.



(a) Error landscape. Errors are scattered uniformly; no clustering in any quadrant.



(b) Precision@k. All three predictors hover near or below the base rate.

Figure 25: PRM error prediction: a null result.

The hypothesis failed. AUCs of 0.54 are functionally chance. PRM_3 performs *worse* than PRM_2 by 0.002.

Error decomposition by failure type (gender margin below threshold, dark activation below threshold, both, or neither) also contradicted the hypothesis: the "OK" quadrant (both signals strong) had the *highest* error rate (10.7%), while "both fail" had zero errors.

Interpretation. The three receptors together explain $R^2 = 0.9815$ of the logit variance. The remaining 1.85% residual is where errors live - but this residual is orthogonal to the receptor subspace by construction. No linear combination of receptor activations can predict which examples fall in the residual's tail. The dark receptor contributes to logit variance through its large mean activation (−205.7), acting as a global calibration term that shifts the entire logit distribution. Removing it disrupts ln_final's normalization for all examples simultaneously, which is why AccDrop = 8.17%. But because the effect is global rather than per-example, it carries no information about individual errors.

This is a clean null result. The dark receptor exists, it matters causally, and its token profile is interpretable (relational-role axis). But it does not function as a routing signal, and the PRM framework does not predict errors on this task.

13 Summary of Findings

Across twelve experiments, the following picture of the GP circuit in GPT-2 Small has emerged.

The circuit runs on three directions. Out of 207 mask-identified write directions, the Mask → CFR → OCA pipeline filters down to exactly 3 that are both causally relevant and mutually independent. These three explain 98.15% of the variance in the model's he-vs-she logit difference ($R^2 = 0.9815$). The remaining 204 directions are ghosts - they appear gender-aligned because they are geometrically correlated with a real receptor, not because they carry independent causal signal. The ghost rate is 93.7%.

Each direction has a distinct computational origin. R1 (L10H9sv0, promotion) is an inherited signal: it is committed to its final sign from the embedding layer itself, before any attention computation. R3 (L9H7sv1, inhibition) is a computed signal: it commits at layer 9, exactly where its host head sits. The dark receptor (L11H1sv1) encodes a relational-role axis (suppressing kinship terms, promoting authority terms) and acts as a global calibration offset with mean activation −205.7 and coefficient of variation 6.2%.

The circuit is coupled, not modular. Cross-talk between R1 and R3 is 41.2% computational (Experiment 2), meaning nearly half the interaction cannot be explained by geometric overlap alone. The coupling is directional: R3 → R1 CompFrac (41.2%) exceeds R1 → R3 CompFrac (19.3%), consistent with the layer ordering (L9H7 fires before L10H9, so R3's perturbation flows through R1's host head but not vice versa).

Errors are dominated by thin margins, not receptor failures. The error taxonomy breaks 219 errors into Type B (inhibition failure, 11%), Type C (both receptors fail, 37%), and Type D (both receptors correct but margin too thin, 52%). Type D is the dominant failure mode, with a median absolute model margin of just 0.89 logit units. The receptor circuit points the right direction but lacks sufficient magnitude to override the 2% residual. Ambiguous names (he-fraction near 0.5) drive most of these thin-margin failures: AUC drops from 0.97 to 0.75 on ambiguous inputs.

The directions generalize across syntax. Receptor AUC remains ≥ 0.92 across all five tested syntactic structures (tag question, because-clause, temporal, prepositional, cross-sentence), with a maximum drop of 5 percentage points. R3 is the most stable across structures.

The dark receptor is real but not predictive. Despite AccDrop = 8.17% (the highest of the three), the dark receptor's AUC for gender classification is 0.526 (chance). Its PRM for error prediction is 0.544 (chance). It contributes to the logit computation as a constant offset, not as a discriminative or gating signal.

14 Next Steps

The most immediate gap in the current results is generalization beyond the GP task. All experiments so far operate on a single task in a single model. The natural next step is to test whether the GP receptor directions transfer to a second task - specifically, the Indirect Object Identification (IOI) task, where the model must distinguish between two names to predict which one is the indirect object. The prediction is that R1 should transfer (since it inherits from the token embedding and the embedding's gender structure is task-independent) while R3 should not (since it is computed by L9H7 for the specific purpose of GP inhibition). A clean dissociation here would validate the inherited-vs-computed distinction as a real structural property of the circuit rather than an artifact of one task.

Before that, a set of analyses on the existing data will characterize the geometry of the causal subspace that CFR and OCA discovered. The 13 CFR survivors can be stacked as rows of a matrix and decomposed via SVD to determine how many independent dimensions they actually span; if the effective rank is 3, the 13 directions live in an approximately 3-dimensional subspace of $\mathbb{R}^{768}$. This connects directly to the superposition hypothesis - the model would be representing 13 partially-redundant gender features in a 3-dimensional causal subspace, with a quantifiable compression ratio. Additionally, projecting the name token embeddings onto R1's direction and R3's direction should reveal whether R1 aligns with the pre-existing gender axis in the embedding space, which would mechanistically explain why R1 is committed from layer 0.

A longer-term direction is to run the full Mask $\rightarrow$ CFR $\rightarrow$ OCA pipeline on a second task entirely (such as factual recall or the greater-than task) and compare the ghost rate, survivor count, and effective rank across tasks. If the pipeline produces similarly sparse, low-rank causal bases on different circuits, the contribution shifts from a case study of GP to a general method for circuit decomposition.